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说话人 1 00:00:00

It's, I think, for the sake of time and maybe to kind of set the stage here, I think most of the comments looks like you're generally on track to address our concerns. And I think the timelines are reasonably aligned for most of these. So I would ask if you can just look at the comments in column j and kind of respond to those when you get a chance. I think, yeah, today, like to focus on maybe these items that we highlighted in orange that we think pose maybe the biggest risk to production schedules and timelines. I think that's the biggest goal of this discussion is to understand which items, if any, may require additional effort or time and kind of align on the path forward.

说话人 2 00:00:43

So that we're all informed on what might impact schedules. Does that make sense? Yeah, that made perfect sense. So yeah, we, and we did quick went through the list and try to like go over everything. But yeah, definitely we can just go over the highlight ones. I know like those are like higher priorities.

说话人 1 00:01:06

Yeah, so maybe I'll just start and like, Po, pose our questions or concerns on these. We can just have a discussion, kind of talk through that. I think again, the rest, these are also important. So I don't want to lose sight of those.

说话人 1 00:01:16

We, it may require additional follow up, but I, yeah, think for today, let's try to focus on if we have time after we can move to the other items as well. So, okay, that's good. I think for this first one, so. It looks like you have committed to having some documentation updated kind of by these dates for each step, which I think is great. I think for our purposes. I think the big question is just understanding the comment here between kind of demo batch run, talking SBPs here in particular, and maybe the cleavage steps. We just wanna make sure we understand what was executed for cleaning, kind of the status of that equipment going into PV and make sure

that's appropriate and then recognize that between PV and when you have to execute your major cleaning, you have a little bit of time to work on the documentation and update as needed and will continue those conversations with, you know, Chris and our other SMEs internally and we can have those discussions. So I think that was kind of the goal here. While we gave, this is. Prior to pre P P Q, not not the intent is not that you're gonna have all your documentation perfect and in place for C major cleaning by pre P P Q start. It's just we wanna understand what has been executed and kind of the status of the equipment going into pre P P Q.

说话人 3 00:02:50

So does that make sense? Yeah, that make sense. So for the cleaning here, I would like to give a quick summary. Okay, so can hear me? Yeah, yes, we can hear. Okay. So in the previous meeting, we talk about, you know, we prepared the cleaning sop and the cleaning record. In each unit operation, let's include the both the cleaning between different batch and be the major cleaning before and after the campaign. So after one week discussion and document preparation.

说话人 3 00:03:32

Until now, we finished the proposal, you know, in the last meeting, we talk about the proposal template. So now, according to the template, we finished the cleaning word cleaning, cleaning sop and cleaning record so we can forward to you for your review. After this meeting. But anyway, if you like, I can share my screen. So the, the, we can have a quick look what the cleaning sop and rocker looks like. But anyway, before the people campaign, the equipment will be. Will be cleaned according to the major cleaning. And after that, we will begin, we will start the PBQ campaign. So what is the status of like the SPP's rig currently? Have you executed a full major cleaning at this point? You know, we. After it was used, we didn't major cleaning. But now, according to our discussion, we talked about, you know, need cleaning procedure and cleaning record. That's not detail enough. So we further updated the cleaning sop and record to make it more clear and more robust.

说话人 1 00:04:58

In their execution. Okay. I think that's the key question or point. Again, for the sake of time, I don't know that we wanna get the too many details on the new cleaning records. Cuz I do wanna go make sure we touch on all these topics that are critical for like discussions on alignment for timeline purposes. But I think what we do need to understand like in the immediate term is just what was executed on SPPS in particular, I guess initially, but what's what will or will be executed on each of the rigs between demo and.

说话人 3 00:05:37

Prior to pre PPQ, because that's probably the most important piece, is that we have confidence and know that the rig is in a pre, that would be, yeah, yeah, that would be major cleaning. After each batch, we will do the extensive cleaning to the, you know, the SPPS unit operation.

说话人 1 00:06:01

That's again after each batches. So again, I think it's good to understand what you're gonna execute after the first pre pq batch. But it's also important to understand if you executed a different cleaning record and your strategy previously was a little bit different, could you just share whatever the document you executed? On the TPS rig in particular, I think if we could see that be maybe helpful to give us a sense of what has been completed. Okay, I can sh quickly share my screen. Can you share with us offline, like send it to us so we can review it internally?

说话人 1 00:06:41

Yes, yes, I can. I think that might be the most efficient thing is that you could just whatever was executed after demo of SPP's, if we could just see that and have like a. Chance to review and maybe if you could just share like summary even of here's what we executed, here's the record. I know we're gonna run a solvent run as well. So that gives us some confidence that the rigs in a good state, but I think that's the.

说话人 1 00:07:02

The biggest question from cleaning going into SPPS, then I think the long term, like you described that you're gonna, you know, enhance your cleaning procedures, SOPs, and that's all. That's great. We'll have ongoing discussions around those before we execute those kind of after pre pbq batches. Yes, it's kind of two, so it's kind of two piece to this. And I think that's maybe the immediate dating activity for pre PQ start is just understanding again what was executed and have confidence the equipments and good status going into pre PQ.

说话人 1 00:07:39

Okay, I've got okay, so I think, okay, so I think if you can share that and we can review and. Again, maybe that cleaning discussion, have further discussion like early next week on that once we've had a chance to review what you've executed, I think there's another set, another action that kind of links to this as well, but I wanna make sure it's really clear. We have a lot of feedback on piping and you know, where there's maybe. Gonna be challenges from a clean perspective, and I know we focused a lot of this around the purification rig, but I do think it's important to understand on other rigs as well, like for SPS and cleavage, you know. For instance, if you have nozzles that are gonna be very high off the tank head and maybe difficult to clean, the spray

balls, want to make sure you've got a robust process for identifying those and making sure that translates into your clean records. So when you go develop those clean records, that's something you call out specifically. And have a robust way to track that it gets cleaned effectively when you do the major cleaning. I think that was one of the follow up items that, again, kind of links to that. So I think that's just to be.

说话人 3 00:08:54

We're here. Yeah, in a update, in a updated cleaning SOP and record, each of the component, they will be identified in a cleaning record. Yeah, this item here. Yeah, so although it's a separate action item, I think this is also one that we wanna make sure as your.

说话人 1 00:09:14

Developing your up enhanced cleaning records that you have a process for capturing those gaps, making sure they get cleaned appropriately, right? Manually or yes, how are you gonna manage that? Okay, so each component and they're identified in a sop and cleaned record.

说话人 3 00:09:30

To assure they're properly cleaned and of acceptable criteria.

说话人 1 00:09:39

So I think that was the cleaning item. So again, I, the reason we have this high highlighted, maybe high risk is just that piece around being confident going into particularly we're gonna start SPPS pre PBQ first here, presumably in a few weeks.

说话人 1 00:09:58

Again, what we need to see is just documentation of what was executed. Enough confidence were. In a good state to start that from a equipment cleaning perspective. And then I think these dates that are reasonable as far as reviewing your enhanced updated documentation, we can work with you on that. So I think the next one that we had highlighted here was around just line sloping. I think so we highlighted this maybe more so because we weren't clear on your proposal here. So if you could maybe expand on this, these different options, wasn't quite clear what this meant, these different options. So I don't know if you could help clarify that.

说话人 2 00:10:45

Hey, Mark. Yes. So basically, so there's, we went through like every equipment and also we went through the like piping, like all of that and try to identify like all the items we need to. App modify

to meet the requirements. And so for these three options, it is like, depends on if there is we ins, we have like CIP or not like F for example, right now, like option 1 to the option 1 right now is like we, we because I know like we know like it's cleaning place is more like robust way to. To have like everything cleaned without any issues. So right now, like a lot like equipment, we do not have the CIP. So for those like new identified like Nami requirements equipments, we like think like long term to have the CIP. So like for first option, as you can see, like if there's, we don't have the CIP and there's some like a few equipments for the connections can it's hard to meet the 2D and like 3D requirements. And because like, can you maybe expand, sorry, interrupt. Just that you say 2D and 3D. That's the, that's what you're referring to there. The nozzle, like the nozzle length. Determine the ratio of the length to diameter. Yeah, yeah, it's okay. Okay. Yeah, that's the 2D and 3D requirement. So I know like from last visit to like T2D is the standard, right? So like we do we have for some like long nozzles. But yeah, and we did decide to like modify them, but like some of them like still like a few equipment still it's a hard to make that happen. So that's why like for option 1, due to the. Also like due to the time, due to the time. So for the first option, we probably still have a few nodules cannot meet the 2 d, three d requirements. That's the option one. Yeah.

说话人 1 00:13:41

So I think maybe just to clarify, I think that assessment is important and that probably ties back to the first item, yeah, to clean. And maybe this last item we just talked about, No. 37, I believe. Yeah, I'll see you. Yeah, what I could say. I think that probably is more addressing this item here, like discussing how you're gonna manage those piping gaps that you can't modify in the short term. I think the intent of this number here, No. 4, was just specifically around a couple of lines that we felt visually looked like they were not sloped appropriately. Oh, okay. So it's very specific to that observation. And it was just maybe a question of if you could just evaluate the line sloping, just kind of confirm that was properly sloped? Generally, in particular for purification or the piping, the higher grade piping specs. So I think that was the intent of this line here. I think what you're describing, like I said, is important but probably ties more to some of the other actions that are specific to the piping modifications in general. Okay, so this was specific. I think if you recall, there were a couple lines that we just visually look like they were.

说话人 3 00:15:11

Maybe not slow, yeah, appropriately. So that, yeah, that was just the follow up to maybe double check those. Okay, general questions. So here I would like to add some comment to here. So at Eastman Camp, no, we're doing the piping work qualification that's include, you know, the proper waste, proper pipeline. And proper slope and proper, whether it's doable. So each interoperation we do the piping work qualification.

说话人 3 00:15:41

Now we're starting from the, you know, from last step to the earlier step, let's call control risk, the criticality of the impact. Generally, I would say if there is some slow problem, it will be either addressed by, you know, add drain, make it drainable or modify the low slope. But anyway, in summary, we're doing the piping work qualification controlled by a protocol. Yeah, so you've already assess the lines you think need to be.

说话人 4 00:16:30

Modified sloping, adjusted. Yeah, the production guys and the engineer guys, they're doing the this work, this piping work. Okay, could you just update then kind of your proposed completion date then to kind of match what works being done and when that, oh, likely be completed. And that evaluation for SPP's included over.

说话人 1 00:16:56

Purification. So this is probably primarily focused on purification, I would say, was the intent. But I, okay, there is kind of a question of just we did not walk through the SPPS equipment set in detail, although line sloping maybe isn't explicitly required there, that if there's excessive like dead. Dead legs or low points that won't drain, that could pose a risk right to the process. So it's worth just evaluating any kind of gross gaps there, observations that would maybe pose a concern. But the primary, I think, concern here was on the actual purification side, making sure that's. Definitely sloped appropriately.

说话人 3 00:17:41

Yeah, we understand the concern. So that's why I said first we do the qualification controlled by protocol. And then if there is some problem, we do either improve the slope or make it drinkable. And the most important, we're doing this work starting from the more critical step, that's from the last step to earlier as PPS step.

说话人 1 00:18:06

That makes sense. I think we were just confuse, again, the confusion was maybe around this column F text. This maybe is discussing maybe a bit of a different topic than what this action was referring to.

说话人 2 00:18:22

Yeah, I think that help. Yeah, very smart. Okay. Yeah, so I also just talk with the team. So, so, so for the purification part, like we definitely can get all the sloping issue resolves prior pre p P Q. And like for the. For the STPs and the cleavage step, like, because it was kind of like categorized as

like C1 and C2, you know, like that's why, eh, like from the beginning we did not like designed like with a sloping, but we. We definitely can double check with that and to identify potential risk and to mitigate that. But for the purification part, we definitely got that done PRI prior to the ppt week. I.

说话人 1 00:19:20

Sorry, I heard maybe, is there any other comments on this item? Or can we move on to the next?

说话人 2 00:19:33

Okay, so, so like our team just like add another comment. So for the like column F, you see the completion date, you know, like this a little bit like later because there's some modifications going on. So they like after the modification, we definitely will recheck the sloping again. To make sure like the sloping align with the requirements. So that's why like the completion dates is kind of like a little bit scary. You know, it's a little bit like a late.

说话人 5 00:20:10

这个是未来，目前。

说话人 2 00:20:12

And even later, so that that's the like basically in the future, if any modification happens, will like that will be the date like to recheck the sloping. But for the current purification steps, the sloping has been checked and make sure they all have like the. Or they all like meets the requirements.

说话人 1 00:20:43

and you think that can be completed before pre- PPQ start for purification.

说话人 2 00:20:53

让我 check，我们那个他问的是，在我们 pre- PPQ 之前，这些制备的 slope 是，都能完事吗？

说话人 5 00:21:08

这些不是几个方案吗？他在选择哪个？

说话人 2 00:21:14

就说，

说话人 5 00:21:18

对，这几个不同的改造方案，决定是否完成。

说话人 2 00:21:22

这个改造其实是，是制备那边的改造，对吧？

说话人 5 00:21:27

对，包含制备那边，比如说一些 BP 网络的更换，还有那些2D的罐口。

说话人 2 00:21:39

嗯，我觉得现在可能有一个 gap 在这。

说话人 5 00:21:46

对。

说话人 2 00:21:48

就我们改造最快也得，那个。

说话人 5 00:21:52

对，12月17日。

说话人 2 00:21:53

哦，那 pre pq 是什么时候？我们知道吗？

说话人 5 00:22:02

这个也还跟他说一下，如果 prequeue 方案一的话，是不影响 prequeue 的结束，不影响后边，目前，按目前最新的计划的结束时间是一致的，启动会晚一点，但是结束时间可以保持一致。

说话人 2 00:22:18

那你就说到置备的时候，他这个 slope 是能没有问题的吗？

说话人 1 00:22:23

Sorry to interrupt.I wonder if, sounds like maybe a little more discussion on.

说话人 2 00:22:29

I think, yeah, so I heard you have a general strategy and maybe there's a little more discussion there just to confirm the status of the piping before pre PDQ on purification. Yeah, so basically for the purification, there's a there's a like, a like modification for the like, for example, to replace with the BPE standard piping. And also for the, like some tanks, we need to modify them to make sure it align with the requirements like, and those modification like may take like that. Like, for example, like there's like three options to. To have that's complete, like after the completion of the modification will ensure like the piping slope is align with requirements. But I think would be helpful. And maybe again, we can talk more on this offline, it sounds like, but it be helpful understand.

说话人 1 00:23:32

What modifications you're planning to make prior to pre PQ start? And then if you're going to go into pre PQ without modifications. This again ties back to that No. 37 item. That's where we need to make sure you've clearly documented where there may be gaps. We're going to accept and acknowledge how you're gonna clean those. Whether it's manual or flow through or inspections, so that we have confidence, we acknowledge the pipings maybe not perfect and meets our standards entirely, but there's a couple of gaps we accept. You have a plan or strategy to capture those in your cleaning records. That's really what happened, I think, so that these kind of tie together.

说话人 1 00:24:15

Some of these items, I think the, yeah, the cleaning documentation and then some of these piping gaps. Okay, again, this item really intended was really to focus on just the sloping of a couple lines. I think what we're getting at here is there's modifications that are gonna be made generally that maybe impact several piping gaps. Yeah.

说话人 2 00:24:35

Yeah, so probably more to come here, but if you can help kind of clarify that, that would be helpful. Yeah, clarification team, like what your strategy or plan is, what what's going to be modified, what's going to be modified later. Yeah, we'll have like a scope of like all the modify equipments and sent to Lily's team. See if that hope that will help.

说话人 1 00:24:58

Yeah, great. If you get, for the sake of time, I'll move to the next item here. That's okay. This next one was just around the CNC. Yeah, the raw material staging area. When you walk through how

you brought that material from the outside into the building, I think what you had here is that. You've committed to a certain date, having a kind of change control to document your plan for renovating the building. Yeah, and I think this item was trying to maybe ask, recognizing that's gonna take some time and you won't probably have that addressed prior to pre pqp, pre pqp start. So the question here was just in the short term, are there any additional like administrative controls we could put in place to try to minimize that risk? So that's kind of the immediate ask for your team to think about. Are there procedural administrative controls, things we could do to just minimize that potential risk of.

说话人 1 00:26:00

Tamination impact. So I think that's, and that probably needs to be considered and then documented. Like in the CCS, we recognize there's a gap. We recognize it'll take some time and maybe difficult without significant time and capital to add an airlock. So what can we do in the short term? I think that's an important question really before we start any pre pqp work. Okay, so there's another action I think that captures the long term. Yep, yep. I don't know. Yeah, so this is kind of short term actions. Okay, think about that and come back with your thoughts on anything we could do to enhance our controls with the current designer setup. Okay, yeah, cuz ultimately we're gonna probably have to accept this risk going into pre PBQ, right? So how do we minimize that impact?

说话人 2 00:27:07

Gotcha. So, so, okay, so basically like a risk assessment, like in, for example, included in the CCS is not.

说话人 1 00:27:16

Enough. We should have like administrative control for the short term. That's what you mean, right? I think we just need to really define the short term mitigation measures. And then, yeah, we probably need to document those in CCS. Ultimately acknowledge that we, the design is not. Does not meet our probably expectations generally, but we think it's reasonable risk to accept. And here's the safeguards put in place or the minute controls put place. You know, but I don't know what those might be. We can talk and brainstorm some thought, you know, collectively some thoughts on what we could do. But just give you a chance to think about that as well. Okay. And again, this is more around just probably physical contamination risk with that outdoor exposure directly into that CNC space. So make sure we're not unintentionally introducing pest risk, you know, pest control management risk or like physical contamination that maybe operators in their clean uniforms get exposed to dust, debris, dirt, bring that into the processing area, things like that. So how. I mean, can we minimize that risk somehow? Okay, so

this is the short term action. There is a longer term action. And I think you're, yeah, thinking about that, which is great. But obviously that will take time to implement. So.

说话人 1 00:28:57

If there's no other questions there, I'll move to the next item that we flagged. I think this is gonna have a lot of discussion here around Passivation. So I wanted to have time to talk about this. So a couple of thoughts here. And I think the recent deviation observation on the activation tank residue ties into this as well. So I wanted, I have a lot of questions I wanna ask.

说话人 1 00:29:22

Right now that maybe we could help clarify or your team could take away and correct me if I misrepresent what I think we understand. So we recognize that there was maybe a lack of passivation on your solvent tanks and solvent headers. I think.

说话人 1 00:29:43

Equally concerning was the observation of a, a weld on the activation header that I believe is a newer section of line that's coming off your bulk solvent header, where you observed a weld that was like rusty or poor quality, is that correct?

说话人 2 00:30:04

Let me double check.

说话人 1 00:30:36

So I think, so is that come from the division from the activation tag? Yes, and I think we see these two is somewhat linked potentially. Hum. So our general concern is we need to have confidence going into. Pre P, P Q that your process insolvent lines are at a good qualified state where they're not going to reduce, you know, metals into the process. So I think obviously passivation is one way to minimize that risk. But now there's probably concerns or questions around the quality of some of the welds.

说话人 1 00:31:19

Hum. And so W1 to ask, maybe to clarify, understand this well, that was observed. First of all, do you have pictures of that you could share? I think that would be really helpful for us to gauge kind of the status of that particular well and the impact. So that'd be one follow up, if you could share a picture of that.

说话人 6 00:31:39

Observation and I maybe I think I did ask for the to take the picture. So if I scan team can recall the pictures we've been taking together on fail check the machine we've been used. I think we do have the picture. Not yet. That be important to add investigation that day. I mean there four weeks ago, that Friday afternoon. I did ask to take the pictures. I'm, I haven't got that being shared. So if you could. Double check with production team or after your information Protection process, maybe you can share with us directly to give us a kind of a directly picture to see because describe too much.

说话人 1 00:32:21

Better than one picture, right. And maybe a follow up to that is just, it'd be good to understand that is a manual weld or like an orbital weld or how that was welded because manual welds maybe inherently would pose more risk if they're not inspected properly or done well. And, and so we could understand that. And I think the obvious question is just do we think this is isolated to this particular weld? Or do you have other manual welds on that header or similar headers, because that may raise questions around the integrity of, you know, those welds. So be good to understand that.

说话人 1 00:33:06

And I think as a follow up, we would like to discuss options around whether we could do some sort of spot check of some of the welds of a similar quality as a confirmation, just to be confident that your equipment's in a good state, good status. So that could be maybe a combination of visual inspections and or boroscoping lines, but I, we probably need to understand what your capability is and what that might look like. But I think that's probably gonna be needed from two fronts. One, to close out that deviation and have a clear sense of the scope of the impact. The other is just this broader question around the metal impurities, I think we still have questions around the Lara project as more clear data supporting the metal catalyzed impurities contributed to that impurity scene investigation on that project.

说话人 1 00:34:05

And then I think even on the two down 7 project, there's still maybe it's not standing question and data we're waiting on to support whether metal catalyzed degradation of depia, for instance, could have led to us see the lot of hide which contributed to this impurity we're seeing. So there's a lot here that ties back to just understanding the quality of your. Your your header and I think we need to think about how do we get the right data to support that we're in a good state to move forward. And I think this is probably an item that could potentially be gating for PV start or pre PQ, pre PQ start. So we need to think about a plan here in a strategy. I would say also if you have data on your solvent tanks, I know you presented maybe a plan to

passivate those and that's pretty lengthy process. I think there's maybe a path forward to try to leverage that data and minimize the scope of work here. But I think we just need to like, you need to share that and kind of define what that looks like.

说话人 3 00:35:09

To help justify that argument. And maybe we can limit the scope to the solvent headers, okay, for only welding and for only metal in a solvent from the speaking, as you said, we have that in place either in TG three site or in our other manufacturing site. For example, in. In Donghua site, the tank farm we used for around 10 years. So every year we pull sample to continuous monitoring the metal in the solvent. I think the data we shared with Ella Lily. So anyway, this is the data from the solvent. But anyway, for the remediation for the tanks and for the header of the welding, we developed a timeline to. You know, for correction. But anyway, we can discuss in a plan, a short term plan and long term plan. For long term plan, of course, we will completely improve the welding and do the persuasion. After that, we can use, but in short term, we can have some discussion how to make the, eh, the, how to say the, the, how to mitigate the risk. For example, we can. We can improve the current filtration to the solvent before use, and in, that's in conjunction with the voluntary data, and then we can plan a proper approach to control the metal in the solvent.

说话人 1 00:37:14

So I think that may be a challenging argument from our end, to be honest up front. We can talk about that, but I think that we need to be able to have confidence that the. Header, and your welds are in a reasonable state that does not introduce significant metal impurities to the process. I don't know that we can just say we're going to filter it out. I mean, we can have that discussion, but I think that'll be a challenging argument with our quality group to justify that.

说话人 1 00:37:45

Especially since we've observed the fact that we observed a weld that we knows rusted, I think that just raises a lot of questions or concerns. So I think what will be helpful is if we can try to minimize the scope as much. Just possible, but have some inspections or spot checks of some similar welds on that line, just to give us a sense of the scope of the issue, if it's localized in one bad weld, that is a much easier, I think, discussion. But if we observe other welds that are potentially. Poor quality or have a rust that raises the concern or risk for the process. I think. So I think that's what we have to understand. Like in the short term, I do agree your tank data, I understand the impact is pretty significant to try to passivate everything. And I'll be clear, passivation probably won't. Address the weld quality. If your welds are poor quality, pass things, probably not going to resolve that. There's kind of two, there's two things here we need to probably think about in general. If the first action is improving the welding and the second is the

passivation. To the working and to the tanks. And I think the tanks you have data. I think we can probably, again, I'm not committing to this now, but I think there's a pathway to have a discussion around leveraging that data. So we don't have to maybe passivate the tanks in the short term necessarily if we have data that supports that, the risk is very low. Okay. I think my question on the piping distribution, I wonder, we ran RS campaign for this process in a different equipment rig. Was any of that solvent header piping common to that equipment set? And I ask because if we can maybe justify that we ran that process without issue using similar solvent headers. Maybe that minimizes the scope of piping question. Maybe we can then focus on the solvent piping that's unique to this equipment set. Yes. Does that make sense as a. Yeah, that make sense for the RS, RS batch, that's the same pipeline and, you know, not distributed from.

说话人 3 00:40:11

From the time of the software distributed from time from. So this should be, you know, this should, you know, relieve the concern. But anyway, we, eh, for the piping, we do the, we will do the preservation.

说话人 1 00:40:31

To mitigate the risk because this is, you know, we can understand this is very important. So I, it'd be helpful if you could share if you have drawings of your solvent header, it'd be good for us to understand which sections of piping, how much of that solvent header is common to what we ran in RS. How much of that is new or unique to what we've run in demo for this process? Cuz that would help us like understand the scope of piping and question. My thought is we can, yeah, if you could share, that'd be really helpful. That might help us just reduce the scope of where we need to focus our efforts potentially.

说话人 3 00:41:14

Okay, we will screen the situation in the of the piping work in ours batch and in the demo batch and PPQ batch and see how many pipeline they're configured the same. So then we can have some more extensive discussion.

说话人 1 00:41:31

That'd be really helpful. And then I think we do need to consider how. So two things before. Pretty good. You start. It'd be good also if you could estimate if you were to do passivation of just the solvent header, not the tank, what does that timeline look like? Does that reduce the timeline significantly to do that work? I know before you quoted a pretty long timeline because you had to do a lot of work with the tank itself. If you take that out of scope, maybe re, re evaluate how long that would take to address the piping. And then maybe the other piece, again for that

investigation of the rest of the stain or residue, but. With regards to the weld, I think we would probably expect some level of either boroscoping or inspection of some welds just to confirm that it's not a systemic issue when it's localized to that one particular weld. Yeah, I think, Lisa, good point. We can really evaluating the timeline.

说话人 3 00:42:38

If we drop the preservation to the tank and only to the preservation to the piping work. At least to me, that would reduce in timeline. Okay. It'd be really good to understand that. I think our team is very eager to understand any of these items that might impact pre PQ starting dates.

说话人 1 00:43:00

And to me, this is probably one that seems like a high risk right now. So I think the sooner we can understand what a reasonable strategy looks like and kind of come up with a plan together, we have a better understanding of like the realistic timeline impact.

说话人 3 00:43:17

Thank you. This is a good discussion and good point. So, oh, we will organize the production and engineer. We can propose another option. If we drop me, you know, the tanks. That would be really helpful. And then again, I think from the investigation on that, well, we're probably going to expect some capas out of that to address.

说话人 1 00:43:41

Well quality, which probably likely we would expect some either boroscoping or slot checking of other manual welds just to ensure those are in good status. And it's maybe an isolated event. So I think that'd be really important as well to gain PB start or pretty good start. Okay, got it. We'll rely on your team to understand kind of scope, what makes sense there. So I again, just you'll have to kind of maybe help us evaluate that, understand what's reasonable. Okay, so I appreciate and this again, I think I see this is one of the higher risk items from like immediate SPPS start, but ultimately, this probably has an impact across the board.

说话人 1 00:44:30

Realistically, there's probably the Lara team, I think has interest in this as well. And it does raise questions that I just challenge team to think about. We happen to observe one weld that was for quality and saw the rust drip into the tank. The question logically is, are there others on other equipment sets? Are we confident? So might be also good to understand your general practice for weld inspections and what was completed? Again, just to, ideally, we get to a point where this is

an isolated event and we can justify that and address it, move on. But if there's a systemic issue, we need to understand that as well.

说话人 3 00:45:16

Okay, I think it's clear after this discussion. And for the piping, the PA, for the piping between the reactors or between equipment or we, the preservation was included, but for the pipeline between the Pang farm and reactors, just now, as with the.

说话人 3 00:45:39

Discussed. We can, you know, we can improve the piping work to include the passivation and we will propose a new timeline. Okay. Okay. So that I think addresses the passivation again. The other piece that's probably equally, it's not more important in my view, is just.

说话人 1 00:46:01

This residue deviation in the well inspections. So they're somewhat linked, but I think they're driven by different observations. So I make sure that's also, but again, the investigation for that purity should drive that activity. And I know, Peter, you're. Maybe in working with the team closely, so you can help maybe drive some of that as well on the site when you're able. So yeah, that will have to be resolved and closed before we can start for PBQ for the SPPS. Okay, so, eh, that was good discussion.

说话人 1 00:46:39

I hope that helps clarify kind of where we're thinking. Certainly be eager to see your thoughts and evaluations on timelines once you had time to digest that. So I see that, again, I see that's a pretty high priority item. So that just, if you can make sure that's a priority on your side as well, that'd be great. There's nothing else there. I think we had one other item maybe at the end. Yeah, two, I guess. Okay. Yeah, one of these is probably a big topic. So I think this one, we talked about the piping, I think with the, I don't know that we need to go in details here directly. I know Andrew and the team are working on the purification side with you closely. So just as you make modifications or propose modifications, you know, let us know if there's questions. The one takeaway, I would say is just if there are. Things you can't modify going into pre TTQ, let's make sure those are captured and then we can address how we ensure those are cleaned as part of your clean strategy. So that's part of the two takeaways there.

说话人 1 00:47:45

And then for the sake of time, this last one, I think. This is probably the one that has probably a lot of discussion as well around the purified water cutter.

说话人 1 00:48:01

Before we go into the observation on the purification side, I have a question I want to make sure to clarify. I think we have the question here. I know we're not using water for the process on SPP's. We have a little bit of water for cleavage. Is there any water off purified water headers that's gonna be used for cleaning on those process steps? For the equipment can mean the water will be involved. Okay, pure for the water. And the question I guess is the observation we made in purification, do you have a similar design gap here for the other process steps? Do you have a long section of piping after the sample point?

说话人 3 00:48:53

Produce those tanks for cleaning. That's the same Si, that's the same situation. And currently is in camp, we're doing the, you know, the piping work from the, you know, from the distribution point. To the use point. According to their, you know, the p p and ID provided by elilili. So the work is ongoing at ECM Cam to address this for all equipment steps.

说话人 3 00:49:35

The work is ongoing. Will that address this for all steps? That's the starting from the more critical steps, from the, eh, from the late steps to the earlier steps. I think the question will be if the gap is similar for P.

说话人 1 00:49:57

SDPS, when we go to use water for cleaning, I think we need to understand the risk because there's similar risk there. If you've got water that is stagnant or hasn't been tested as part of that loop. Again, we maybe can accept that risk, but we need to just document that and justify why it's okay.

说话人 3 00:50:23

Even though it's not process water, even though we're using it for cleaning, there still is microbial contamination risks, right? Yes. Internally we have some discussion. Yeah, yeah, internally we have some discussion for the early steps. First, we can improve our current practice. You know, specify the requirement of drink, be drink af, drink the pipeline after use, and also use f fresh host before the before. Collection between the use point to the distribution point. And secondly, we can propose a short term plan and long term plan. For example, for the pipico batch and or for the demo batch, how we how we do the how we plan the use of pure flow water.

说话人 1 00:51:22

And before and after PPQ bet, before commercial bet, how we do the work or in propose a different proposal.Yeah, yeah, I think it'd be great to see that and understand.

说话人 1 00:51:34

Again, I think that's the right approach.I recognize you may not be able to modify the piping before pre PPQ.So I think we just need to see that approach and just align on that strategy, make sure.We've considered the risk and have that documented.So this is also, I see probably again, the risk may be lower for cleaning, but I think it still is a risk we need to document and be aware of and align on accepting that risk going into pre PBQ.And then obviously for purification, it's maybe a bit of a higher risk being process water.So the same thing there will have to just document our strategy, short term, long term, and talk through what that looks like.Yeah, I'll be able to do it.

说话人 1 00:52:16

But okay, go ahead.Yeah, I just want to make sure the early steps were on your radar as well.As far as the water used for cleaning, I know we focused a lot of discussion on purification.But it sounds like you're already thinking about that, which is great.So I think we need to see for all steps and all water uses, kind of that what you discussed, a strategy, short term, long term, and just a risk assessment to make sure we're aligned on the path forward.

说话人 3 00:52:45

Yeah, correct.At least if we follow this approach, we're, you know, this is under control according to the risk management.And you know, we also do the correction according to the risk from the critical step to the next critical step.

说话人 1 00:53:07

So that's, you know, that's a key point of the risk management in GMP.Yeah, fully agree.I, I, again, I think that for the immediate, just with the SPPS starting soon, probably seeing that risk assessment would be needed before we start. Just so everybody's aligned on.Recognizing the risk in the strategy in the short term, while you're still working on the long term, addressing the higher risk steps downstream as well.But I think that don't want to lose sight of that.I guess that's gonna be something we can make sure is in front of people so we can review that.Okay, I've got.So after this meeting, I will organize the, you know, the approach.For the water from the critical step to the less critical step, and then make a short term action and a long term action in different states.Okay, yeah, that'd be very helpful.Thank you.Okay, thank you.

说话人 1 00:54:11

So I think the last one highlighted here, I don't know, Andrew, if you're on, I guess it's a maybe purification specific item. I mean, it was all steps, I guess. I think our concern on this was maybe just the timeline. We flag this as a pre ppq item.

说话人 3 00:54:38

This ties back to the water, I guess. So that's maybe, yeah, this, so this ties back to the water. Yes, this is the short term. So I think this is kind of tied to what we're just talking about. Yeah, for this topic, we can also included in there, you know, the strategy and risk management.

说话人 1 00:54:59

You know, use of the pure for the water. Perfect. I think that makes sense. Yeah, great. I think, okay, so good. We're done. Timer. So I think those are the, again, I don't want to discredit that there's a lot of effort and work here on both sides. I appreciate your team's effort and kind of reviewing our comments, and. In general, I think a lot of these are on track, which is great.

说话人 1 00:55:23

We tried to highlight again the high risk items that we discussed. So we'll send some notes as well from the meeting. Just we've been running copilot the background. So that way there's a clear list of actions and what needs to be followed up on. So appreciate the discussion. And again, I probably will need more discussion on these other items potentially that maybe are less of a risk timelines. So I don't want to discredit that those are all important, but these are the five or six that we highlighted that we want to make sure we're working on a active plan here to mitigate risk.

说话人 1 00:55:56

So thank you for the discussion. And I think it's really helpful to have that dialogue. So there any other last minute questions from your side or any additional follow UPS that you.

说话人 3 00:56:07

You wanna discuss here in the four minutes we have left. So by tomorrow, we will, according to our plan, we will send you the, you know, the detailed work for cleaning management. Include the improved protocol, improved SOP, and improved record. After your review, if you think it's, we need another meeting for discussion, we're more than happy to have a meeting maybe tomorrow or the day after tomorrow.

说话人 1 00:56:44

And I know the purification JPT, I think it's still scheduled for Friday. So maybe an opportunity for that team in particular to pull some of these purification specific topics out, discuss if that is needed. Maybe we'll wait to see your feedback.

说话人 1 00:57:00

Back on some of the discussion today, and then we can kind of determine when the next logical meeting needs to be to follow up. Again, I think the urgency on our end is just making sure we understand what actions are ongoing that might gate pre PQ start so we can have a realistic understanding of the timelines. So I think that's probably the first step here, and then the obvious will be ongoing discussion as you're working through these and implementing these actions.

说话人 1 00:57:31

Okay, we've got and it looks like the chats maybe got some of the, yeah, I just do the notes. So some of the notes are in the chat here. So if your team could reference these, I think this is. Just the AI in the background compiling our discussion. So hopefully this summarize things. Accurately. And if you have questions, let us know. We'll do the same offline and view this as well. So internally, we will provide the bit limited, right?

说话人 1 00:58:01

Yeah, thank you so much again for the, yeah, this is the, yeah, I'm sure notice this is really helpful discussion. So please reach out with any additional questions before we meet again. So thank you. Okay, thank you. Have a good evening. Thank you all. Bye, everyone.

说话人 3 00:58:16

Okay, thanks.